

**Notes: Chen, Hwang, and Peng (RFS, 2025)**  
**Why Do Investors Like Short-leg Securities? Evidence from a Textual Analysis  
of Buy Recommendations**

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# 1 Big picture

- **Question.** Why do investors like the stocks that sit in the *short leg* of cross-sectional return anomalies, that is, stocks that trade at relatively high prices and earn unusually low future returns?
- **Core difficulty.** Traditional return data alone usually cannot distinguish among competing theories. Low future returns for short-leg securities are consistent with at least three broad stories:
  - **Safety story:** investors rationally value short-leg securities as safe-haven assets and accept lower returns because these stocks hedge bad states of the world.
  - **Supremacy / exuberance story:** investors over-extrapolate recent success or become overconfident about favorable signals, so they view these firms as unusually strong and bid prices above fundamentals.
  - **Lottery story:** investors are attracted by upside potential and pay a premium for the small chance of an extreme positive payoff.
- **Paper’s move.** Instead of trying to infer beliefs from prices alone, the authors read what market participants actually *say*. They parse written buy recommendations in analyst reports and Seeking Alpha (SA) articles and classify whether the language emphasizes safety, supremacy, or lottery-like features.
- **Main finding.** Across anomalies, text sources, investor-specific word lists, and multiple robustness checks, the dominant explanation is **lottery / upside potential**. The paper’s headline conclusion is that perceived upside potential plays a material role in driving investor demand for short-leg securities.
- **Validation.** The authors then run a one-time survey of institutional and retail investors with actual long positions in short-leg securities. The survey-based evidence lines up closely with the text-based evidence.
- **Contribution.** The paper proposes text as a scalable complement to investor surveys. Surveys are more precise, but text is cheap, retroactive, and easy to apply at large scale.
- **Why it matters.** The paper reinterprets a classic asset-pricing question through investor discourse. Its message is that any successful theory of the cross-section of expected returns must explain investors’ strong preoccupation with the possibility of extreme positive outcomes.

## 2 Data and measurement (key objects)

### 2.1 Three conceptual frameworks

The paper organizes the question around three frameworks.

- **Traditional risk / safe-haven framework.** Investors may willingly pay a high price for stocks that protect them in bad states, such as low consumption growth, disaster states, or periods of high macro uncertainty.

- **Irrational beliefs / supremacy framework.** Investors may over-extrapolate past good news or be overconfident about private information, causing them to see some stocks as unusually strong, dominant, or guaranteed to keep doing well.
- **Lottery framework.** Investors may be drawn to stocks that seem like gambles: most likely not exceptional, but with a small chance of a spectacular payoff.

The empirical design is built to ask which of these three narratives appears most often in actual buy recommendations.

## 2.2 Survey-based word lists

The paper constructs dictionaries from investor surveys rather than imposing ad hoc word lists.

### 2.2.1 Institutional-investor word lists

- In October 2022, the authors survey **100 U.S.-based institutional investors** through CoreData Research.
- Respondents are wealth, mutual fund, pension fund, and hedge fund managers who actively invest in U.S. stocks.
- The key survey prompts ask investors to list words they would use to describe:
  - a safe-haven stock,
  - a stock that has been doing well and is expected to continue doing well,
  - a stock that is somewhat of a gamble with enormous upside if it pays off.
- The five most common base terms are:
  - **Safety:** conservative, defensive, protection, reliable, stable.
  - **Supremacy:** competitive, expanding, leader, outperformer, strong.
  - **Lottery:** gamble, potential, speculative, upside, volatile.
- After expanding to business-relevant word forms, the final institutional dictionaries contain **18 safety words, 19 supremacy words, and 35 lottery words.**

### 2.2.2 Retail-investor word lists

- In November 2022, the authors survey **303 U.S. retail investors** through Prolific.
- The five most common retail base terms are:
  - **Safety:** reliable, safe, secure, stable, steady.
  - **Supremacy:** consistent, excellent, growth, innovative, winner.
  - **Lottery:** exciting, gamble, potential, speculative, volatile.
- The final retail dictionaries contain **29 safety words, 42 supremacy words, and 40 lottery words.**

## 2.3 Text sources

### 2.3.1 Sell-side analyst reports

- Source: Investext.
- After excluding algorithmic or non-stock-specific providers, the sample retains **664 brokerages**.
- The paper processes reports from the **top 100 brokerages**, which account for **98%** of all analyst reports.
- Final sample: **1,718,353 analyst reports** from 2006–2021.
- Recommendation breakdown:
  - **1,032,917 buy recommendations**,
  - 625,232 hold recommendations,
  - 60,204 sell recommendations.
- Average report length is **1,146 words**.

### 2.3.2 Seeking Alpha (SA) articles

- Source: articles in the *stock ideas* section of Seeking Alpha.
- Sample restrictions: single-ticker articles, at least 50 words, 2006–2021.
- Final sample: **141,746 articles**.
- Tag breakdown:
  - **73,061 long ideas** (treated as buy recommendations),
  - 9,530 short ideas,
  - 59,155 untagged.
- Average SA long-idea length is **1,217 words**.

## 2.4 Anomaly data and the definition of short-leg securities

- The paper builds on Chen and Zimmermann’s open-source anomaly database.
- The starting point is **205 firm-level predictors** of returns.
- For each firm characteristic and each month, stocks are ranked and the **bottom decile** is defined as the **short leg**.
- For practical coverage reasons, the final text analysis uses **186 characteristics**. The missing cases are mostly anomalies whose short-leg securities have too little analyst or SA coverage, often because those stocks are very small.
- The paper also groups the 205 predictors into **32 economic categories**, such as valuation, profitability, R&D, default risk, volatility, and information proxy.

## 2.5 Validation survey of actual investors

- Institutional validation survey: **450 institutional investors**, fielded in March–April 2024.
- Retail validation survey: **314 U.S. retail investors**, fielded in June 2024.
- Respondents are asked to think about up to eight stocks they bought in the past year and choose the *primary* reason for each purchase: safety, supremacy, or lottery.
- Because of data availability and realized short-leg purchases, the survey validation can be performed for **53 characteristics** in the institutional sample and **48 characteristics** in the retail sample.

## 2.6 Important measurement caveat

A stock being described with lottery language does *not* pin down a single deep mechanism. It may reflect:

- probability weighting in cumulative prospect theory, or
- overly optimistic beliefs about the right tail of the return distribution.

So throughout the paper, the empirical result is best interpreted as: investors are unusually drawn to short-leg securities for their **perceived upside potential** / **lottery-like features**, without the text data fully separating preferences from tail optimism.

# 3 Models and empirical specifications (all equations + interpretation)

## 3.1 Classification of recommendations by anomaly-month

Fix a firm characteristic  $i$  and month  $t$ .

- Rank stocks on characteristic  $i$ .
- Define the **short-leg securities** as the bottom decile in month  $t$ .
- Define **short-leg recommendations** as buy recommendations published in month  $t$  on those short-leg securities.
- Define **other recommendations** as all other buy recommendations in month  $t$ .

This procedure is repeated separately for each of the 186 characteristics.

## 3.2 Text shares

For a given text document  $d$  with total word count  $N_d$ , define:

$$\text{Safety}(d) = \frac{\#\{\text{safety words in } d\}}{N_d}, \quad \text{Supremacy}(d) = \frac{\#\{\text{supremacy words in } d\}}{N_d},$$

$$\text{Lottery}(d) = \frac{\#\{\text{lottery words in } d\}}{N_d}.$$

These are scaled word-frequency measures. The paper also accounts for simple negation by checking whether one of a set of negation words appears within three words before a dictionary term.

### 3.3 Main abnormality statistic

For each anomaly  $i$ , the main statistic compares the average word share in short-leg recommendations to the average in other recommendations. For safety, the paper uses

$$\Delta\text{Safety}(\%) = \frac{\text{Safety}(\%)_{\text{short-leg recs}} - \text{Safety}(\%)_{\text{other recs}}}{\text{Safety}(\%)_{\text{other recs}}}. \quad (1)$$

Analogously,

$$\begin{aligned} \Delta\text{Supremacy}(\%) &= \frac{\text{Supremacy}(\%)_{\text{short-leg recs}} - \text{Supremacy}(\%)_{\text{other recs}}}{\text{Supremacy}(\%)_{\text{other recs}}}, \\ \Delta\text{Lottery}(\%) &= \frac{\text{Lottery}(\%)_{\text{short-leg recs}} - \text{Lottery}(\%)_{\text{other recs}}}{\text{Lottery}(\%)_{\text{other recs}}}. \end{aligned}$$

**Decision rule.** For each anomaly  $i$ :

- if only one of the three  $\Delta$ 's is positive and statistically significant at the 5% level, that perception explains investors' liking of short-leg securities for anomaly  $i$ ;
- if more than one is positive and significant, the perception with the *largest relative difference* is the one that *primarily* explains the attraction;
- if none is positive and significant, the result is **inconclusive**.

### 3.4 Two alternative scalars

To ensure results are not driven by a tiny denominator in equation (1), the paper also uses two alternatives:

$$\Delta\text{Safety}(\%) = \frac{\text{Safety}(\%)_{\text{short-leg recs}} - \text{Safety}(\%)_{\text{other recs}}}{\text{Safety}(\%)_{\text{short-leg recs}} + \text{Safety}(\%)_{\text{other recs}}}, \quad (2)$$

$$\Delta\text{Safety}(\%) = \text{Safety}(\%)_{\text{short-leg recs}} - \text{Safety}(\%)_{\text{other recs}}. \quad (3)$$

The analogous changes are applied to supremacy and lottery. The paper finds that the dominance of the lottery explanation survives all these alternative scalings.

### 3.5 Validation-survey analogue

The survey exercise mirrors the text design.

For each anomaly  $i$ , define short-leg stocks as the stocks purchased by a respondent that lay in the short leg during the relevant survey window. Let "other stocks" be purchased stocks that never lay in the short leg for anomaly  $i$ .

Then compute, for example,

$$\Delta\text{Safety}^{\text{survey}}(\%) = \frac{\Pr(\text{safety reason} \mid \text{short-leg stock}) - \Pr(\text{safety reason} \mid \text{other stock})}{\Pr(\text{safety reason} \mid \text{other stock})},$$

with analogous expressions for supremacy and lottery.

The same significance and “primary explanation” rules are then applied.

### 3.6 Economic-category analysis

The paper next maps the 186 characteristics into 32 economic categories and repeats the same classification exercise category by category. This lets the authors ask not just “what is the average explanation across all anomalies?” but also “for what kinds of anomalies do safety, supremacy, or lottery matter most?”

### 3.7 Theory-driven self-defined word lists

As a robustness exercise, the authors construct a more theory-driven set of safety, supremacy, and lottery words. The purpose is to check that the results are not an artifact of the survey-generated dictionaries.

### 3.8 BERT / FinBERT classification

The paper also replaces dictionary counts with a sentence-level classification method based on transformer models.

- Three binary classifiers are trained: safety, supremacy, and lottery.
- The base language model is **FinBERT**, a BERT model pretrained on financial text.
- Each sentence is classified as relevant or irrelevant for each category.

This is a much less transparent but more context-sensitive robustness check.

### 3.9 Long-leg sell recommendations

The paper also tries to explain why investors *dislike* long-leg securities using sell recommendations and negated language. This extension is conceptually interesting, but empirically weaker because sell recommendations are much rarer, especially in SA. The paper therefore treats the short-leg buy-recommendation evidence as the central result.

## 4 Identification and instruments (what makes the estimates credible)

This paper is not an IV paper in the usual causal-treatment sense. Its identification challenge is instead: *how can one distinguish among competing behavioral explanations for low future returns?*



## 4.1 Why standard observational data are not enough

A stock earning low future returns after trading at a high price can be explained by at least three broad mechanisms:

- rational demand for safety,
- exuberant or overconfident beliefs,
- lottery preference or perceived upside potential.

Because all three can generate high prices and low subsequent returns, prices alone do not identify the underlying narrative.

## 4.2 What the text design identifies

The paper identifies which narrative is *most unusually present in written justifications*. The logic is:

- if short-leg recommendations disproportionately use safety language relative to other recommendations, safety is a plausible explanation;
- if they disproportionately use supremacy language, exuberance / overconfidence is a plausible explanation;
- if they disproportionately use lottery language, upside potential is a plausible explanation.

So the object identified is not a structural preference parameter; it is the **dominant expressed rationale** attached to short-leg securities.

## 4.3 Why analyst reports and SA articles are informative

The paper's maintained assumption is that the stories analysts and SA authors choose to tell partially reflect investor beliefs. The economic logic is a **catering / confirmation-bias** argument:

- readers prefer narratives that resonate with their own belief systems;
- authors who want attention and readership tailor their stories to those belief systems;
- therefore, the stories that appear disproportionately in recommendations reveal something meaningful about investor perceptions.

This is especially plausible because the paper uses two distinct text environments:

- analyst reports, interpreted as more institutional,
- SA articles, interpreted as more retail.

Finding similar patterns across both environments materially strengthens the argument.

## 4.4 Why the validation survey is powerful

The strongest credibility check is the direct survey of investors with actual long positions in short-leg securities. This survey asks the same question the text analysis is trying to infer indirectly.

If the text-based and survey-based answers line up, then the core assumption behind the text design becomes much more believable.

## 4.5 Remaining limitations

- The text method is noisy.
- Lottery language does not cleanly separate prospect-theory preferences from optimistic tail beliefs.
- The paper identifies broad categories well, but finer-grained distinctions within categories remain difficult.
- For some anomaly categories, the text evidence is inconclusive because coverage is sparse or language differences are weak.

## 4.6 Relation to the literature

The paper sits at the intersection of three literatures:

- **Investor surveys:** it complements direct surveys by offering a cheaper, scalable alternative.
- **Financial textual analysis:** instead of measuring merely positive vs. negative tone, it studies *why* investors like a stock.
- **Anomaly / asset-pricing literature:** it offers a new way to think about the short leg of return anomalies.

# 5 Tables and figures

## 5.1 Table 1: survey-based word lists

- Table 1 is foundational because it maps abstract theories into measurable text objects.
- It shows the institutional and retail base terms and the expanded word-form lists.
- The table also makes clear that “lottery” is operationalized through words such as *gamble*, *potential*, *speculative*, *upside*, and *volatile*.

## 5.2 Internet Appendix Figures IA5 and IA6: sample texts

These examples are useful because they let the reader see the classification idea directly.

- **Safety example:** RBC Bearings is pitched as an attractive stock “in times of uncertainty given its defensive nature.”

- **Supremacy example:** Ansys is described as “best-in-class” and a continuing growth leader.
- **Lottery example:** Magnum Hunter is described as a “purely speculative play” with upside potential.

These examples are especially helpful for remembering the economic difference between the three categories.

**Table 1**  
Survey-based word lists

| Safety words                                                                                                                                                                                                                                               | Supremacy words                                                                                                                                                                                                                                                                                                                                                                                              | Lottery words                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>A. Institutional investors' survey-based word lists</i>                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                            |
| conservative, defensive, protection, reliable, stable                                                                                                                                                                                                      | competitive, expanding, leader, outperformer, strong                                                                                                                                                                                                                                                                                                                                                         | gamble, potential, speculative, upside, volatile                                                                                                                                                                                                                                                                                                                                                                           |
| defensively, protected, protections, reliability, reliabilities, reliability, stabilities, stableness, stableness, stably                                                                                                                                  | competitively, expand, expanded, expands, expansion, expansions, lead, leaders, leadership, leaderships, leading, leads, led, strongly                                                                                                                                                                                                                                                                       | gambled, gambler, gamblers, gambles, gambling, gambings, potentialities, potentiality, potentially, potentials, speculate, speculated, speculates, speculating, speculation, speculations, speculatively, speculativeness, speculateness, speculator, speculators, upsidess, volatiles, volatilities, volatility, volatilizable, volatilize, volatilized, volatilizes, volatilizing                                        |
| <i>B. Retail investors' survey-based word lists</i>                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                            |
| reliable, safe, secure, stable, steady                                                                                                                                                                                                                     | consistent, excellent, growth, innovative, winner                                                                                                                                                                                                                                                                                                                                                            | exciting, gamble, potential, speculative, volatile                                                                                                                                                                                                                                                                                                                                                                         |
| reliability, reliabilities, reliability, safeness, safenesses, secured, securely, secureness, securenesses, secures, securing, stability, stabilities, stableness, stablenesses, stably, steadied, steadies, steadily, steadiness, steadinesses, steadying | consistence, consistencies, consistency, consistencies, consistently, excel, excelled, excellence, excellences, excellently, excelling, excels, grew, grow, grower, growers, growing, growths, grown, grows, growths, innovate, innovated, innovates, innovating, innovation, innovations, innovational, innovativeness, innovativenesses, innovator, innovators, win, wins, winners, winning, winnings, won | excite, excited, excitement, excitements, excites, excitingly, gambled, gambler, gamblers, gambles, gambling, gambings, potentiality, potentialities, potentially, potentials, speculate, speculated, speculates, speculating, speculation, speculations, speculatively, speculativeness, speculateness, speculator, speculators, volatiles, volatility, volatilizable, volatilize, volatilized, volatilizes, volatilizing |

This table reports our survey-based safety, supremacy, and lottery words. We first report our five base terms, followed by all of the word forms of the base terms. We describe how we arrive at the base terms and construct the corresponding word forms in [Section 2.2](#).

**Table 2**  
Why do investors like short-leg securities? Text-based evidence

|                                                            | Fraction of times investors explain [primarily explain] their liking of short-leg securities through |                         |                       |                  |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------|-----------------------|------------------|
|                                                            | Perceived safety (1)                                                                                 | Perceived supremacy (2) | Perceived lottery (3) | Inconclusive (4) |
| <i>A. Institutional investors' survey-based word lists</i> |                                                                                                      |                         |                       |                  |
| Sell-side analyst reports                                  | 12% [8%]                                                                                             | 29% [17%]               | 68% [55%]             | 20%              |
| SA articles                                                | 7% [6%]                                                                                              | 16% [11%]               | 57% [54%]             | 30%              |
| <i>B. Retail investors' survey-based word lists</i>        |                                                                                                      |                         |                       |                  |
| Sell-side analyst reports                                  | 24% [5%]                                                                                             | 27% [18%]               | 69% [65%]             | 12%              |
| SA articles                                                | 10% [8%]                                                                                             | 23% [13%]               | 63% [57%]             | 22%              |

This table reports the frequency with which a particular reason is used to explain the buy recommendation of a stock that resides in the short leg of an anomaly. For each of 186 firm characteristics, we conduct the following analysis: Each month, we rank stocks based on the firm characteristic. The literature indicates that stocks in the lowest decile (“short-leg securities”) trade at relatively high prices and earn unusually low future returns. We term the analyst reports and SA articles that recommend buying these short-leg securities as “short-leg recommendations,” and all other buy recommendations as “other recommendations.” The analysts and SA contributors may recommend buying a short-leg security for at least three reasons: (1) They believe that the stock adds safety to the overall portfolio as it exhibits stability, consistency, and resilience during economic downturns or adverse conditions (“Perceived safety”). (2) They are very confident that something good will happen to the company, or the stock has been doing well and is expected to continue to do well (“Perceived supremacy”). (3) While risky and uncertain, the stock has the potential for quick, substantial gains (“Perceived lottery”). To examine whether the short-leg recommendations emphasize a stock’s perceived safety, supremacy, or lottery-like feature, we compute the average fraction of safety words, *Safety* (%), in the analysts’ (or SA contributors’) short-leg recommendations. To determine whether the use of safety words is abnormally high, we also compute the average *Safety* (%) across the analysts’ (or SA contributors’) other recommendations. We then calculate the difference between the former and the latter (on a relative basis),  $\Delta\text{Safety}$  (%), and test whether this difference is positive and statistically significant at the 5% level. We conduct similar tests for  $\Delta\text{Supremacy}$  (%) and  $\Delta\text{Lottery}$  (%). If  $\Delta\text{Safety}$  (%) is positive and statistically significant at the 5% level, we conclude that one reason investors like the short-leg securities is the perceived safety. Similarly, if  $\Delta\text{Supremacy}$  (%) or  $\Delta\text{Lottery}$  (%) are positive and significant, we conclude that investors like the short-leg securities for their perceived supremacy or lottery-like feature, respectively. If none of the differences are statistically significant at the 5% level, we deem our result “Inconclusive.” If more than one of the differences is positive and significant, we label the rationale associated with the largest relative difference as the one that “primarily explains” investors’ liking of the short-leg securities. We perform this analysis for each of the 186 firm characteristics and, ultimately, calculate the fraction of times investors explain [primarily explain] their liking of short-leg securities through the stocks’ perceived safety, supremacy, or lottery-like feature. We also compute how many times our results are inconclusive. Panel A reports the fractions based on the institutional investor word list shown in [Table 1](#), while panel B reports the fractions based on the retail investor word list, also listed in [Table 1](#).

## 5.3 Table 2: main text-based evidence

Table 2 is the core empirical table.

### Panel A: institutional-investor word lists.

- **Sell-side analyst reports:**
  - safety explains short-leg attraction in 12% of cases and is the *primary* explanation in 8%;
  - supremacy explains it in 29% of cases and is primary in 17%;
  - lottery explains it in 68% of cases and is primary in 55%;
  - 20% of cases are inconclusive.
- **SA articles:**
  - safety explains 7% and is primary in 6%;
  - supremacy explains 16% and is primary in 11%;
  - lottery explains 57% and is primary in 54%;

– 30% are inconclusive.

**Panel B: retail-investor word lists.** The same qualitative result holds. Lottery remains the dominant explanation:

- analyst reports: primary explanation = 65% lottery,
- SA articles: primary explanation = 57% lottery.

**Takeaway.** No single rationale explains every anomaly, but perceived upside potential is the dominant rationale overall.

## 5.4 Table 3: evidence by economic category

| Economic category<br>(Number of anomalies in category)<br>(The one or top two anomalies in category) | Sell-side analyst reports                                                                        |                               |                             |                     | SA articles                                                                                      |                               |                             |                     |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------|---------------------|--------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------|---------------------|
|                                                                                                      | Fraction of times<br>investors primarily explain their liking<br>of short-leg securities through |                               |                             |                     | Fraction of times<br>investors primarily explain their liking<br>of short-leg securities through |                               |                             |                     |
|                                                                                                      | Perceived<br>safety<br>(1)                                                                       | Perceived<br>supremacy<br>(2) | Perceived<br>lottery<br>(3) | Inconclusive<br>(4) | Perceived<br>safety<br>(5)                                                                       | Perceived<br>supremacy<br>(6) | Perceived<br>lottery<br>(7) | Inconclusive<br>(8) |
| Accounting [5] (e.g., Accruals, Abnormal accruals)                                                   | 0%                                                                                               | 20%                           | 40%                         | 40%                 | 0%                                                                                               | 0%                            | 60%                         | 40%                 |
| Asset composition [6] (e.g., Net up. assets, Cash & assets)                                          | 0%                                                                                               | 25%                           | 50%                         | 25%                 | 0%                                                                                               | 0%                            | 25%                         | 25%                 |
| Cash flow risk [3] (Cash flow to price variance)                                                     | 0%                                                                                               | 0%                            | 100%                        | 0%                  | 0%                                                                                               | 0%                            | 100%                        | 0%                  |
| Corporate account [2] (e.g., Piotroski) income, included expenses                                    | 20%                                                                                              | 20%                           | 40%                         | 20%                 | 0%                                                                                               | 0%                            | 80%                         | 0%                  |
| Default risk [1] (D-score)                                                                           | 0%                                                                                               | 0%                            | 100%                        | 0%                  | 0%                                                                                               | 0%                            | 100%                        | 0%                  |
| Earnings event [2] (Ret. over earn. announcement period, decline in coverage)                        | 0%                                                                                               | 0%                            | 50%                         | 50%                 | 0%                                                                                               | 0%                            | 50%                         | 50%                 |
| Earnings forecast [5] (e.g., Earnings forecast to stock price, Predicted analyst forecast errors)    | 0%                                                                                               | 0%                            | 80%                         | 20%                 | 0%                                                                                               | 0%                            | 100%                        | 0%                  |
| Earnings growth [3] (e.g., Average earn. growth, Stock in run, upturn)                               | 0%                                                                                               | 0%                            | 67%                         | 33%                 | 0%                                                                                               | 0%                            | 67%                         | 33%                 |
| External financing [1] (e.g., Indicator of initial public offering, Common up. liquidation)          | 0%                                                                                               | 10%                           | 82%                         | 0%                  | 0%                                                                                               | 0%                            | 82%                         | 18%                 |
| Information proxy [2] (From size)                                                                    | 100%                                                                                             | 0%                            | 0%                          | 0%                  | 100%                                                                                             | 0%                            | 0%                          | 0%                  |
| Investment [10] (e.g., CAPEX to revenues, R equity to assets)                                        | 0%                                                                                               | 17%                           | 72%                         | 0%                  | 0%                                                                                               | 11%                           | 67%                         | 17%                 |
| Investment growth [3] (e.g., CAPEX growth over time, CAPEX growth over past three yrs.)              | 0%                                                                                               | 0%                            | 100%                        | 0%                  | 0%                                                                                               | 0%                            | 100%                        | 0%                  |
| Lead lag [9] (e.g., Economic link momentum, Price change coefficient)                                | 22%                                                                                              | 11%                           | 11%                         | 56%                 | 11%                                                                                              | 0%                            | 22%                         | 67%                 |
| Leverage [4] (e.g., Book leverage, Mkt. leverage)                                                    | 25%                                                                                              | 0%                            | 25%                         | 50%                 | 0%                                                                                               | 0%                            | 25%                         | 75%                 |
| Liquidity [9] (e.g., Bid-ask spread, Bid ask spread)                                                 | 22%                                                                                              | 0%                            | 67%                         | 11%                 | 11%                                                                                              | 0%                            | 88%                         | 22%                 |
| Long-term reversal [9] (e.g., Long-run reversal, Momentum reversal)                                  | 0%                                                                                               | 83%                           | 17%                         | 0%                  | 0%                                                                                               | 50%                           | 33%                         | 17%                 |
| Momentum [9] (e.g., Momentum (12-month), Momentum (6-month))                                         | 0%                                                                                               | 0%                            | 67%                         | 33%                 | 0%                                                                                               | 0%                            | 50%                         | 44%                 |
| Option risk [2] (Volatility smirk, Put volatility minus call volatility)                             | 0%                                                                                               | 0%                            | 50%                         | 50%                 | 0%                                                                                               | 0%                            | 100%                        | 0%                  |
| Other [25] (e.g., GDM governance index, IPO age)                                                     | 12%                                                                                              | 28%                           | 44%                         | 16%                 | 8%                                                                                               | 20%                           | 40%                         | 32%                 |

(Continued)

| Economic category<br>(Number of anomalies in category)<br>(The one or top two anomalies in category) | Sell-side analyst reports                                                                        |                               |                             |                     | SA articles                                                                                      |                               |                             |                     |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------|---------------------|--------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------|---------------------|
|                                                                                                      | Fraction of times<br>investors primarily explain their liking<br>of short-leg securities through |                               |                             |                     | Fraction of times<br>investors primarily explain their liking<br>of short-leg securities through |                               |                             |                     |
|                                                                                                      | Perceived<br>safety<br>(1)                                                                       | Perceived<br>supremacy<br>(2) | Perceived<br>lottery<br>(3) | Inconclusive<br>(4) | Perceived<br>safety<br>(5)                                                                       | Perceived<br>supremacy<br>(6) | Perceived<br>lottery<br>(7) | Inconclusive<br>(8) |
| Ownership [5] (e.g., Active shareholders, Number of inst. owners)                                    | 0%                                                                                               | 0%                            | 33%                         | 67%                 | 0%                                                                                               | 0%                            | 0%                          | 100%                |
| Payout indicator [1] (First month when consistent div. payments start to pay div.)                   | 0%                                                                                               | 0%                            | 0%                          | 100%                | 0%                                                                                               | 0%                            | 0%                          | 100%                |
| Profitability [9] (e.g., Gross profits to total assets, Net inc. to book equity)                     | 0%                                                                                               | 11%                           | 89%                         | 0%                  | 0%                                                                                               | 11%                           | 89%                         | 0%                  |
| R&D [4] (e.g., Advertising expense, R&D expense to mkt. value)                                       | 0%                                                                                               | 70%                           | 0%                          | 25%                 | 0%                                                                                               | 70%                           | 25%                         | 0%                  |
| Recommendation [2] (Analyst consensus rating)                                                        | 0%                                                                                               | 0%                            | 0%                          | 100%                | 0%                                                                                               | 0%                            | 0%                          | 100%                |
| Risk [6] (e.g., CAPM beta, Coherence rating monthly net.)                                            | 17%                                                                                              | 33%                           | 33%                         | 17%                 | 33%                                                                                              | 0%                            | 67%                         | 0%                  |
| Sales growth [6] (e.g., Revenue growth rate, Sales growth over inventory growth)                     | 0%                                                                                               | 17%                           | 50%                         | 33%                 | 0%                                                                                               | 0%                            | 17%                         | 83%                 |
| Short sale disposition [5] (e.g., SR, RO, and forecast dispersion, SR, RO, and bid. volatility)      | 0%                                                                                               | 0%                            | 20%                         | 80%                 | 0%                                                                                               | 0%                            | 20%                         | 80%                 |
| Short-term reversal [1] (Short-term reversal)                                                        | 0%                                                                                               | 0%                            | 100%                        | 0%                  | 0%                                                                                               | 0%                            | 100%                        | 0%                  |
| Size [1] (Size)                                                                                      | 100%                                                                                             | 0%                            | 0%                          | 0%                  | 100%                                                                                             | 0%                            | 0%                          | 0%                  |
| Valuation [15] (e.g., Book-to-mkt. ratio, Total assets to mkt. value)                                | 7%                                                                                               | 27%                           | 67%                         | 0%                  | 7%                                                                                               | 13%                           | 73%                         | 7%                  |
| Volatility [5] (e.g., Mkt. risk based on CAPM, Mkt. risk based on Fama)                              | 0%                                                                                               | 0%                            | 100%                        | 0%                  | 0%                                                                                               | 0%                            | 100%                        | 0%                  |
| Volume [5] (e.g., Lagged 2-month trading volume, Monthly trading volume trend)                       | 20%                                                                                              | 0%                            | 80%                         | 0%                  | 0%                                                                                               | 20%                           | 60%                         | 20%                 |

Chen and Zimmerman (2022) divide all "anomalies" into 32 distinct "economic categories." This table reports the results from panel A in Table 2, broken down into these 32 economic categories. To give a sense of which anomalies reside in a particular economic category, we list the top one or top two anomalies within each category based on the Google Scholar citations of the corresponding academic papers as of June 2022. Columns (1)–(4) display the fraction of anomalies where sell-side analysts primarily explain their liking of short-leg securities based on perceived safety, supremacy, and lottery. Column (4) reports the number of instances in which our findings are inconclusive. Columns (5)–(8) display the corresponding numbers for the SA contributors. We bold all cases in which both the fraction based on sell-side analysts and that based on the SA contributors are greater than 60%. CAPEX = capital expenditure; CAPM = capital asset pricing model; div. = dividends; earn. = earnings; FFA = Fama and French three-factor model; GDM = Governance, Skills, and Metrics; Mkt. = idiosyncratic; inc. = income; incl. = industry; inst. = institutional; RO = institutional ownership; IPO = initial public offering; mkt. = market; op. = operating; ret. = returns; SR = short interest ratio; yrs. = years.

Table 3 refines the main result by economic category.

- **Information proxy** anomalies are primarily explained by **safety**.
- **R&D** and **long-term reversal** anomalies are primarily explained by **supremacy**.
- **Lottery** is the dominant explanation for many important categories, including:
  - cash flow risk,
  - default risk,
  - earnings forecasts,
  - earnings growth,
  - external financing,
  - investment,
  - investment growth,
  - profitability,
  - short-term reversal,
  - valuation,
  - volatility.

A few especially memorable lines from Table 3:

- **Default risk:** 100% lottery in both analyst and SA results.
- **Volatility:** 100% lottery in both analyst and SA results.
- **Information proxy:** 100% safety in both analyst and SA results.
- **Profitability:** 89% lottery in both analyst and SA results.

## 5.5 Firm-size sensibility check

The paper also checks whether the pattern makes economic sense.

- If smaller stocks are more naturally viewed as lottery-like, then lottery should matter more when short-leg securities are small.
- This is exactly what the paper finds: when short-leg securities are relatively small, the analyst text points to lottery **78%** of the time versus **31%** when they are relatively large.
- Conversely, supremacy matters more for larger short-leg securities: **28%** for large stocks versus **6%** for small stocks.

## 5.6 Table 4: survey-based validation

Table 4 is the most important validation exercise.

- **Institutional investors' actual purchases:**
  - safety = 17%,
  - supremacy = 13%,
  - lottery = 64%,
  - inconclusive = 6%.
- **Retail investors' actual purchases:**
  - safety = 6%,
  - supremacy = 10%,
  - lottery = 81%,
  - inconclusive = 2%.
- **Agreement with text-based method:**
  - analyst reports vs. institutional survey = **65.2%**,
  - SA articles vs. retail survey = **91.2%**.

This is strikingly strong support for the text approach.

**Table 4**  
Why do investors like short-leg securities? Survey-based evidence

|                                                                                         | Fraction of times investors [primarily explain]<br>their actual purchase of short-leg securities through |                               |                             |                     |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------|---------------------|
|                                                                                         | Perceived<br>safety<br>(1)                                                                               | Perceived<br>supremacy<br>(2) | Perceived<br>lottery<br>(3) | Inconclusive<br>(4) |
| Institutional investors                                                                 | [17%]                                                                                                    | [13%]                         | [64%]                       | 6%                  |
| Proportion agreement (Sell-side analyst reports; Institutional investor survey) = 65.2% |                                                                                                          |                               |                             |                     |
| Retail investors                                                                        | [6%]                                                                                                     | [10%]                         | [81%]                       | 2%                  |
| Proportion agreement (SA articles; Retail investor survey) = 91.2%                      |                                                                                                          |                               |                             |                     |

This table presents the responses from institutional and retail investors regarding their primary motivation for purchasing short-leg securities. In March/April 2024, we survey 450 institutional investors, asking them to reflect on the stocks they purchased over the past year. If an investor bought more than eight stocks, we instruct them to consider the first eight that come to mind. For each stock, we ask whether their key motivation for the purchase was the stock's perceived safety, supremacy, or lottery. In June 2024, we ask the same questions to 314 U.S. retail investors. For each of the 186 firm characteristics  $i$ , we then conduct the following analysis. Similar to Table 2, we define "short-leg securities" as the stocks that our survey participants report having bought and that reside in the short leg based on firm characteristic  $i$  at any point between April 2023 and March 2024 for the institutional investors and between July 2023 and June 2024 for the retail investors. We label the stocks that our participants purchased but that never resided in the short leg as "other stocks." For each firm characteristic with the necessary data, we compute  $\Delta\text{Safety}(\%)$ , which measures the fraction of times the institutional investors report having bought the short-leg securities for their perceived safety relative to how often the same investors cite safety as the key reason for purchasing other stocks. We then test whether this difference is positive and statistically significant at the 5% level. We conduct similar tests for  $\Delta\text{Supremacy}(\%)$  and  $\Delta\text{Lottery}(\%)$ . If  $\Delta\text{Safety}(\%)$  is positive and significant, we conclude that our survey participants disproportionately purchased short-leg securities for their perceived safety. Similarly, if  $\Delta\text{Supremacy}(\%)$  or  $\Delta\text{Lottery}(\%)$  is positive and significant, we conclude that our participants disproportionately purchased short-leg securities for their perceived supremacy or lottery-like features, respectively. If none of the differences are statistically significant at the 5% level, we classify the result as inconclusive. If more than one difference is statistically significant, we assign the perception with the largest relative difference as the primary explanation for why our survey participants bought the short-leg securities. We perform this analysis for each firm characteristic and, ultimately, calculate the fraction of times the investors primarily explain their actual purchases of short-leg securities through perceptions of safety, supremacy, or lottery. We also compute how frequently our results are inconclusive. Below, we report the fractions separately for the institutional and the retail investors. Additionally, we compute the percentage of cases in which our text-based inferences of why investors primarily favor short-leg securities (from Table 2) and our survey-based inferences (from this table) are the same.

**Table 5**  
Why do investors like short-leg securities? Sensitivity analyses

|                                                        | Fraction of times investors explain [primarily explain]<br>their liking of short-leg securities through |                               |                             |                     |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------|---------------------|
|                                                        | Perceived<br>safety<br>(1)                                                                              | Perceived<br>supremacy<br>(2) | Perceived<br>lottery<br>(3) | Inconclusive<br>(4) |
| <i>A. Alternate denominator</i>                        |                                                                                                         |                               |                             |                     |
| Sell-side analyst reports                              | 12% [8%]                                                                                                | 29% [17%]                     | 68% [55%]                   | 20%                 |
| SA articles                                            | 7% [6%]                                                                                                 | 16% [11%]                     | 57% [54%]                   | 30%                 |
| <i>B. Numerator only</i>                               |                                                                                                         |                               |                             |                     |
| Sell-side analyst reports                              | 14% [9%]                                                                                                | 31% [18%]                     | 65% [49%]                   | 24%                 |
| SA articles                                            | 6% [4%]                                                                                                 | 21% [10%]                     | 59% [54%]                   | 31%                 |
| <i>C. "Theory-driven" self-defined word lists</i>      |                                                                                                         |                               |                             |                     |
| Sell-side analyst reports                              | 17% [12%]                                                                                               | 27% [13%]                     | 66% [58%]                   | 16%                 |
| SA articles                                            | 12% [10%]                                                                                               | 18% [11%]                     | 56% [53%]                   | 25%                 |
| <i>D. BERT</i>                                         |                                                                                                         |                               |                             |                     |
| Sell-side analyst reports                              | 18% [16%]                                                                                               | 26% [10%]                     | 68% [62%]                   | 12%                 |
| SA articles                                            | 13% [10%]                                                                                               | 16% [10%]                     | 61% [59%]                   | 21%                 |
| <i>E. High Google Scholar citations anomalies only</i> |                                                                                                         |                               |                             |                     |
| Sell-side analyst reports                              | 13% [13%]                                                                                               | 34% [17%]                     | 70% [57%]                   | 13%                 |
| SA articles                                            | 6% [6%]                                                                                                 | 21% [13%]                     | 64% [60%]                   | 21%                 |
| <i>F. Data since publication only</i>                  |                                                                                                         |                               |                             |                     |
| Sell-side analyst reports                              | 10% [8%]                                                                                                | 28% [18%]                     | 68% [54%]                   | 19%                 |
| SA articles                                            | 8% [6%]                                                                                                 | 16% [11%]                     | 56% [53%]                   | 30%                 |
| <i>G. Older anomalies only</i>                         |                                                                                                         |                               |                             |                     |
| Sell-side analyst reports                              | 13% [11%]                                                                                               | 33% [13%]                     | 73% [60%]                   | 16%                 |
| SA articles                                            | 9% [7%]                                                                                                 | 20% [11%]                     | 64% [62%]                   | 20%                 |

This table reports the frequency with which investors use a particular reason to explain their buy recommendations of stocks that reside in the short leg of an anomaly. The analyses are identical to those in panel A of Table 2, except that we now report results based on a different denominator and based on the numerator only as detailed in Section 4.1 (panels A and B); apply our self-defined word lists as detailed in Section 4.3 (panel C); apply BERT rather than compute the fractions of safety, supremacy or lottery words as detailed in Section 4.4 (panel D); consider only the firm characteristics whose corresponding academic paper's Google Scholar citation as of June 2022 is in the top quartile (panel E); consider only the data since the corresponding academic paper was published (panel F); or consider only the firm characteristics whose corresponding academic paper was published in the first quartile of our sample period (panel G).

## 5.7 Table 5: robustness and alternative methods

Table 5 shows that the core message is robust.

- **Alternative scalars:** lottery still dominates.
- **Self-defined theory-driven word lists:** analyst primary lottery = 58%, SA primary lottery = 53%.
- **BERT / FinBERT:** analyst primary lottery = 62%, SA primary lottery = 59%.
- **High-citation anomalies only:** analyst primary lottery = 57%, SA primary lottery = 60%.
- **Data since publication only:** analyst primary lottery = 54%, SA primary lottery = 53%.
- **Older anomalies only:** analyst primary lottery = 60%, SA primary lottery = 62%.

The exact percentages move somewhat, but the ranking of the three explanations does not.

## 5.8 Long-leg sell recommendations

The paper also studies sell recommendations on long-leg securities.

- Dictionary methods perform poorly because sell recommendations are scarce and negated safety / supremacy / lottery terms are rare.
- BERT does better, but the results are much weaker than for short-leg buy recommendations.
- This asymmetry is consistent with the theory emphasis on *overpricing* of short-leg securities, rather than underpricing of long-leg securities.

## 6 Short summary

- The paper asks what beliefs draw investors to short-leg securities.
- It compares three narratives: safety, supremacy, and lottery.
- It uses investor-generated word lists to parse 1.7 million analyst reports and 141,746 Seeking Alpha articles.
- For each anomaly, it measures whether short-leg buy recommendations unusually emphasize safety, supremacy, or lottery relative to other buy recommendations.
- Main result: **lottery / upside potential is the dominant rationale**. In the main text-based results, it is the primary explanation in 55% of analyst-report cases and 54% of SA cases using the institutional-investor word lists.
- Survey validation of actual investors points in the same direction: lottery is primary in 64% of institutional-investor cases and 81% of retail-investor cases.
- The pattern is robust to alternative word lists, alternative scalars, BERT/FinBERT classification, and subsample restrictions.
- The paper’s deepest message is that investor attraction to the right tail of the return distribution is central for understanding why short-leg securities become overpriced and earn low future returns.

## 7 Conclusion

### 7.1 Core conclusion (what the paper establishes)

The paper establishes that investors’ written justifications for buying short-leg securities are most often rooted in **perceived upside potential** rather than perceived safety or general supremacy. This conclusion survives multiple measurement choices and aligns closely with direct survey evidence from investors who actually hold such stocks.

### 7.2 Economic intuition

The simplest way to remember the result is:

- short-leg securities are expensive;
- expensive securities must be liked by someone;
- when people explain *why* they like them, they most often talk about *upside*.

So the paper shifts the emphasis away from the purely rational safe-haven story and toward a world in which investors are unusually attracted to the possibility of big wins.

### 7.3 What this means for anomaly interpretation

A useful implication is that not all anomaly short legs are the same.

- Some look like safe assets.
- Some look like “great firms” that investors are overly enthusiastic about.
- But many of the most important short legs are discussed in language that looks much more like speculative upside.

That is why the category-level analysis is valuable: it shows where each mechanism is most plausible.

### 7.4 Methodological takeaway

This paper is also important methodologically. It shows that textual analysis can be used not just to measure tone, but to classify competing economic narratives. The method is not as precise as direct surveys, but its scale, low cost, and historical reach make it powerful as a complementary research tool.

The one-sentence takeaway:

Short-leg securities seem to be overpriced not mainly because investors view them as especially safe, but because investors are disproportionately attracted to their perceived upside potential and lottery-like features.